

Mobius iFlex 2000 L Bioreactor - 400-2000 L working volume

merck-iflex-2000

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	400 L	Source-supported
Maximum working volume	2000 L	Source-supported
Nominal vessel / bag size	2000 L	Source-supported
Total vessel volume loaded into the form	2400 L	Source-supported
Initial working volume loaded into the form	400 L	Derived / estimated
Product-contact material	Ultimus single-use film	Source-supported
Height : diameter	1.64	Derived / estimated
Internal / chamber diameter	1.16 m	Source-supported
Liquid height at maximum working volume	1.9 m	Derived / estimated
Impeller / mixing type	Pitched-blade turbine	Source-supported
Impeller count	1	Source-supported
Impeller diameter	0.406 m	Source-supported
Power number	3.7	Source-supported
Initial agitation	55 rpm	App assumption
Maximum agitation	102 rpm	Source-supported
Initial aeration	0.05 vvm	App assumption
Maximum aeration	0.3 vvm	App assumption
Baffled vessel	Yes	Source-supported
Maximum pressure	0.02757903 bar(g)	Source-supported
Default operating pressure	0 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 400 L

Source-supported | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7; General System Specifications

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 400 L.

Limitations: Locally archived MK_DS12340EN Ver. 4.0, March 2025 confirms previously gathered values. Four-blade down-pumping pitched impeller, bottom mounted 15 degrees from centre; X-baffle. Metric dimensions take precedence over rounded inch equivalents.

Local original: [Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 \(March 2025\) - PDF page 7](#)

Maximum working volume - 2000 L

Source-supported | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7; General System Specifications

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2000 L.

Limitations: Locally archived MK_DS12340EN Ver. 4.0, March 2025 confirms previously gathered values. Four-blade down-pumping pitched impeller, bottom mounted 15 degrees from centre; X-baffle. Metric dimensions take precedence over rounded inch equivalents.

Local original: [Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 \(March 2025\) - PDF page 7](#)

Nominal vessel / bag size - 2000 L

Source-supported | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7; General System Specifications

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2000 L.

Limitations: Locally archived MK_DS12340EN Ver. 4.0, March 2025 confirms previously gathered values. Four-blade down-pumping pitched impeller, bottom mounted 15 degrees from centre; X-baffle. Metric dimensions take precedence over rounded inch equivalents.

Local original: [Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 \(March 2025\) - PDF page 7](#)

Total vessel volume loaded into the form - 2400 L

Source-supported | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7; General System Specifications

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2400 L.

Limitations: Locally archived MK_DS12340EN Ver. 4.0, March 2025 confirms previously gathered values. Four-blade down-pumping pitched impeller, bottom mounted 15 degrees from centre; X-baffle. Metric dimensions take precedence over rounded inch equivalents.

Local original: [Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 \(March 2025\) - PDF page 7](#)

Initial working volume loaded into the form - 400 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: min(400 L, 2000 L) = 400 L

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Ultimus single-use film

Source-supported | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 4; Bioreactor Flexware assemblies

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Ultimus single-use film.

Limitations: Film brand confirmed; woven nylon reinforcement and Irgafos 168-free contact layer described. This does not identify all contact-layer polymers. Page 7 stainless steel is support hardware, not the bag film.

Local original: [Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 \(March 2025\) - PDF page 4](#)

Height : diameter - 1.64

Derived / estimated | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7; General System Specifications

Rule: $H = 4 V / (\pi D^2)$; liquid $H/D = H / D$

Selection / calculation: $V = 2 \text{ m}^3$; $D = 1.158 \text{ m}$; $H = 1.898991389381315 \text{ m}$; $H/D = 1.639888937289564$

Limitations: Derived liquid aspect ratio. Does not reuse an overall vessel ratio or generic profile ratio as if it described the liquid.

Local original: Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7

Internal / chamber diameter - 1.16 m

Source-supported | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7; General System Specifications

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1.16 m.

Limitations: Locally archived MK_DS12340EN Ver. 4.0, March 2025 confirms previously gathered values. Four-blade down-pumping pitched impeller, bottom mounted 15 degrees from centre; X-baffle. Metric dimensions take precedence over rounded inch equivalents.

Local original: Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7

Liquid height at maximum working volume - 1.9 m

Derived / estimated | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7; General System Specifications

Rule: $H = 4 V / (\pi D^2)$; liquid $H/D = H / D$

Selection / calculation: $V = 2 \text{ m}^3$; $D = 1.158 \text{ m}$; $H = 1.898991389381315 \text{ m}$; $H/D = 1.639888937289564$

Limitations: Ideal-cylinder estimate from published diameter and working volume. Ignores tapered walls, curved bottoms and internals; not a manufacturer liquid-height specification.

Local original: Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7

Impeller / mixing type - Pitched-blade turbine

Source-supported | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7; General System Specifications

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Pitched-blade turbine.

Limitations: Locally archived MK_DS12340EN Ver. 4.0, March 2025 confirms previously gathered values. Four-blade down-pumping pitched impeller, bottom mounted 15 degrees from centre; X-baffle. Metric dimensions take precedence over rounded inch equivalents.

Local original: Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7

Impeller count - 1

Source-supported | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 9; Mixer M201 notes

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1.

Limitations: Mixer is used with a single-use impeller; single bottom-mounted design also described on pages 5 and 7.

Local original: Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 9

Impeller diameter - 0.406 m

Source-supported | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7; General System Specifications

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.406 m.

Limitations: Locally archived MK_DS12340EN Ver. 4.0, March 2025 confirms previously gathered values. Four-blade down-pumping pitched impeller, bottom mounted 15 degrees from centre; X-baffle. Metric dimensions take precedence over rounded inch equivalents.

Local original: Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7

Power number - 3.7

Source-supported | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7; General System Specifications

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 3.7.

Limitations: Locally archived MK_DS12340EN Ver. 4.0, March 2025 confirms previously gathered values. Four-blade down-pumping pitched impeller, bottom mounted 15 degrees from centre; X-baffle. Metric dimensions take precedence over rounded inch equivalents.

Local original: [Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 \(March 2025\) - PDF page 7](#)

Initial agitation - 55 rpm

App assumption | Large mammalian-vessel scaling rule

Rule: $\text{baseRpm} = \text{maxVolume} \geq 1000 \text{ L} ? 55 : 80$

Selection / calculation: 2000 L is at least 1000 L -> 55 rpm

Limitations: The simulator lowers its generic starting agitation at larger scale. This is an internal scale heuristic, not a manufacturer limit. This is the initial agitation setting loaded into the form.

Maximum agitation - 102 rpm

Source-supported | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 9; Mixer M201, Operating Process Range

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 102 rpm.

Limitations: Published process range 27-102 rpm, not the wider instrument range. The app stores the maximum only; the lower limit is retained here.

Local original: [Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 \(March 2025\) - PDF page 9](#)

Initial aeration - 0.05 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: $\text{PROFILE_DEFAULTS}["\text{mammalian}"].\text{baseVvm} = 0.05$

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 0.3 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: $\text{PROFILE_DEFAULTS}["\text{mammalian}"].\text{maxVvm} = 0.3$

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - Yes

Source-supported | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 7; General System Specifications

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Yes.

Limitations: Locally archived MK_DS12340EN Ver. 4.0, March 2025 confirms previously gathered values. Four-blade down-pumping pitched impeller, bottom mounted 15 degrees from centre; X-baffle. Metric dimensions take precedence over rounded inch equivalents.

Local original: [Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 \(March 2025\) - PDF page 7](#)

Maximum pressure - 0.02757903 bar(g)

Source-supported | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 9; PE001 bag pressure: Operating Process Range and MFC-stop interlock

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.02757903 bar(g).

Limitations: 0.4 psi above atmospheric pressure, converted using $1 \text{ psi} = 0.06894757293168 \text{ bar}$. Used as the simulator warning ceiling. This is the published process/interlock limit, NOT a vessel design-pressure rating or the 0-6 psi sensor span. Detection tolerance is $\pm 0.05 \text{ psi}$.

Local original: Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 9

Default operating pressure - 0 bar(g)

App assumption | Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 9; PE001 process range and atmospheric zero adjustment

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 0 bar(g).

Limitations: Atmospheric app starting point replaces the legacy 0.05 bar setting, which exceeds the documented process/interlock limit. Zero is not a manufacturer-recommended culture setpoint.

Local original: Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025) - PDF page 9

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: $\text{maxOxygenFraction} = 100$

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

These presets are primarily mammalian-cell-culture archetypes.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

Published 2:1 is TOTAL vessel height/diameter, not liquid H/D; the liquid geometry remains estimated. No 200/2000 L table values are extrapolated to 50 or 1000 L.

AN12431EN (July 2023), Table 1 p. 2, corroborates working range, vessel diameter, pitched-blade geometry and power number. The March 2025 dedicated datasheet remains preferred. PG12163EN Rev. 2 p. 4 gives 21.1 cm for the 200 L impeller versus the selected 21.0 cm in the 2025 datasheet; do not mix editions silently.

Geometry caution: AN12431EN p. 2 prints minimum/maximum liquid-height ratios greater than one; PG12163EN p. 4 repeats this. A reversed ratio heading is plausible but unconfirmed. The maximum-fluid/total-height ratio 0.8 is nominal, not an absolute liquid-height measurement. Neither ratio overwrites the existing liquid-height calculation. Validation scope and conflicts: sources/AN12431-REVIEW.md.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Merck Mobius iFlex - MK_DS12340EN Ver. 4.0 (March 2025)

merck-iflex-ds12340en-v4-2025-3e5f3a2ab18a.pdf

SHA-256: 3e5f3a2ab18af511e560342b4f7e5f41b6b8b7183a2e13a185e1dbb2de3a5a44